ParEGO Implementation Fix and Fixed-Budget Performance

How does the PHYSBO ParEGO implementation fix affect optimization performance, and how does corrected ParEGO compare with HVPI, NDS and Random under a fixed evaluation budget?

PHYSBO 3.2.1 | VLMOP2 | 20 seeds | 50 evaluations per run | Local revision 2026-10-16

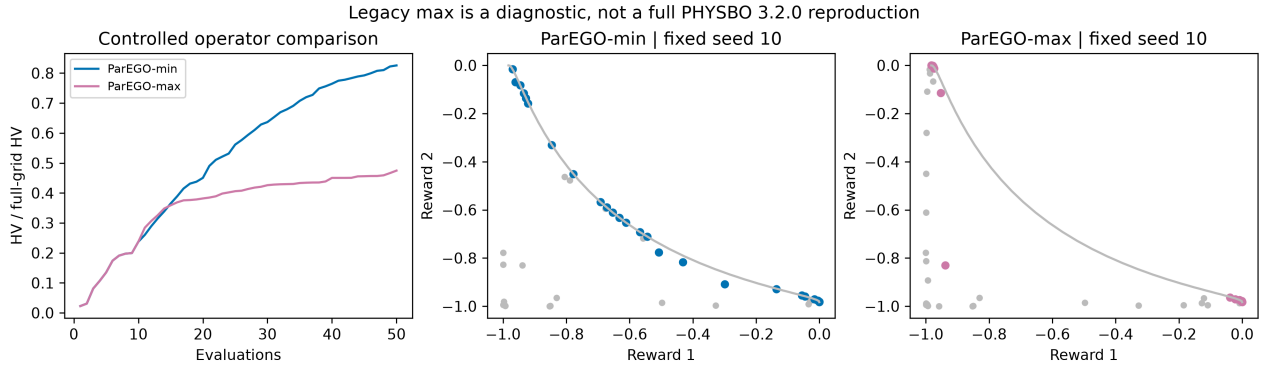
Both experiments use PHYSBO 3.2.1 and the fixed VLMOP2 two-objective function on a 101 x 101 grid over [-2,2]^2, with reference (-1,-1). Each trajectory has 50 total evaluations including ten shared random initial candidates. Seeds are 10-29; exact GP is used and retrained every ten probes. Hypervolume (HV) is the dominated trade-off area relative to the fixed reference; larger is better. Full-grid HV is approximately 0.334518. The 90% target is approximately 0.301066. The full grid is evaluated only for offline reporting.

Experiment 1 - Effect of the implementation fix

Experiment 1 changes only the weighted Chebyshev reduction inside PHYSBO 3.2.1: diagnostic max versus corrected min. Initial points, normalization, GP settings and an independently logged weight stream are paired. For weights (0.5,0.5), extreme points (-1,0) and (0,-1) score -0.525 under min and -0.025 under max; balanced point (-0.4,-0.4) scores -0.220 under both. The ranking changes, followed by different search trajectories.

Controlled operator	Final raw HV mean [95% interval]
ParEGO-max	0.1588 [0.1341, 0.1830]
ParEGO-min	0.2762 [0.2629, 0.2876]

At 50 evaluations, the paired raw-HV difference (min minus max) is 0.1174 [0.0891, 0.1469]. The corrected reduction improves mean final quality in this controlled experiment. The official fix is upstream PHYSBO work. This is an operator-only diagnostic within 3.2.1, not an end-to-end comparison of complete software releases.



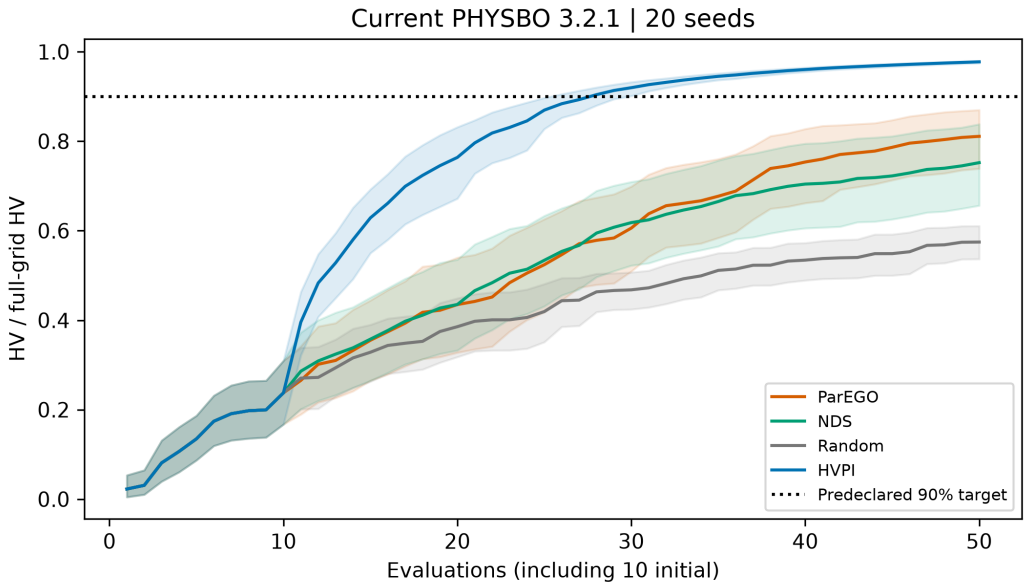
The corrected diagnostic ParEGO-min mean in Experiment 1 is 0.2762, whereas native ParEGO in Experiment 2 is 0.2713. These are separate conditions with different weight mechanisms: the diagnostic uses an independent logged paired weight stream; native ParEGO uses the library mechanism. Their records must not be pooled or treated as the same branch.

The study has two experiments: controlled fix effect, then the corrected native strategy comparison.

Experiment 2 - Fixed-budget strategy comparison

Experiment 2 uses the corrected native library strategies. Native ParEGO has mean final raw HV 0.2713, compared with 0.3270 for HVPI, 0.2516 for NDS and 0.1922 for Random. The 90% target is reached in 4/20, 20/20, 6/20 and 0/20 runs, respectively. Thus the fix improves controlled ParEGO behavior, while native ParEGO still has lower observed mean final HV than HVPI at the same evaluation cap. ParEGO has a higher observed mean than NDS and Random; no new pairwise significance test is claimed.

Strategy	Final raw HV mean [95% interval]	90% target hits	Decision seconds
ParEGO	0.2713 [0.2472, 0.2910]	4/20	2.718
HVPI	0.3270 [0.3260, 0.3278]	20/20	6.948
NDS	0.2516 [0.2195, 0.2804]	6/20	2.650
Random	0.1922 [0.1795, 0.2043]	0/20	<0.001



Query-budget summaries - mean raw HV

Queries	ParEGO	HVPI	NDS	Random
10	0.0795	0.0795	0.0795	0.0795
20	0.1455	0.2555	0.1456	0.1290
50	0.2713	0.3270	0.2516	0.1922

Target sensitivity - successes out of 20

Grid-HV target	ParEGO	HVPI	NDS	Random
80%	15/20	20/20	13/20	0/20
90%	4/20	20/20	6/20	0/20
95%	0/20	20/20	0/20	0/20

The conclusions concern this VLMOP2 grid, 20 seeds, fixed model settings and evaluation cap. The existing 10/20/50-query summaries and 80%/90%/95% targets describe budget and target dependence within the same runs. Pointwise 95% quality intervals use 10,000 percentile seed-bootstrap draws with seed 20261015. Success denominators retain all runs. This evidence does not establish a universal ranking, a win in every seed, or physical experimental performance. Algorithm time is a separate local computation measurement.

Retained evidence: 120 original trajectories / 6,000 evaluations. Verify with python scripts/run_upgrade.py --verify-only. Original per-run values were not changed or rerun. Source links, provenance, attribution and reproduction details are in docs/REPORT.md.